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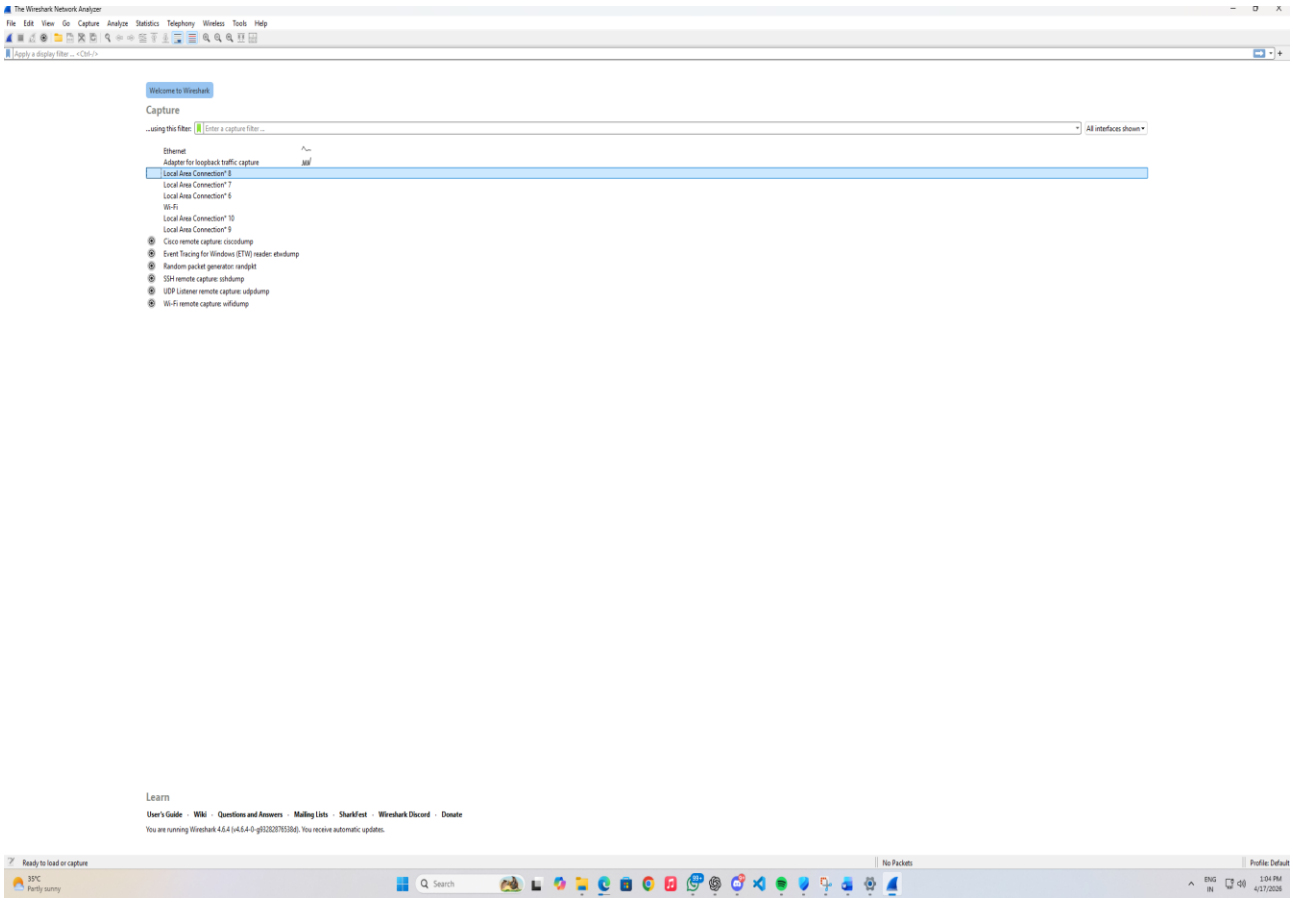
Assignment-1

1. **To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.**

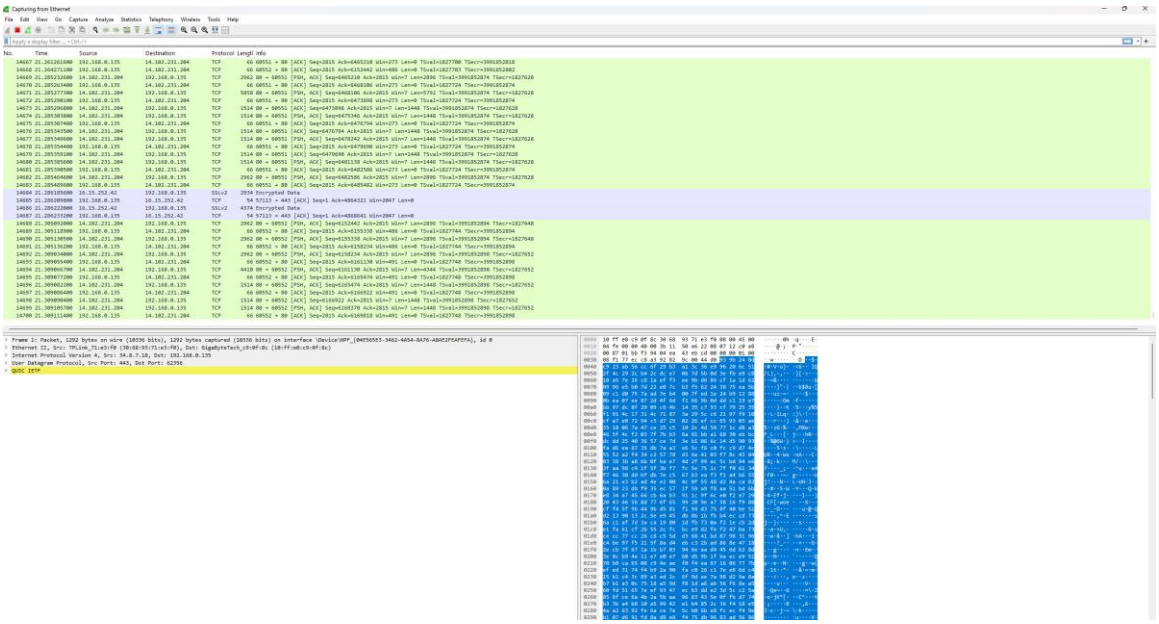
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
    - Wi-Fi
    - Ethernet



Step 2: Start Packet Capture

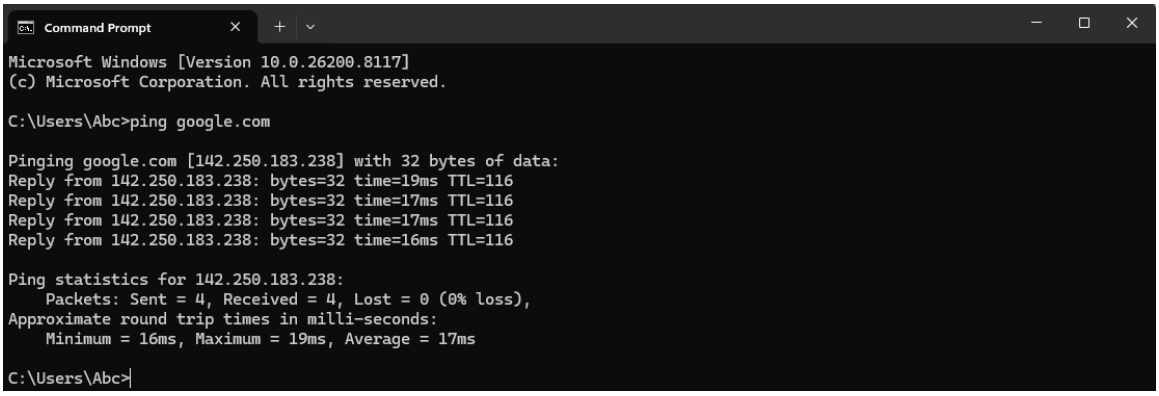
- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com



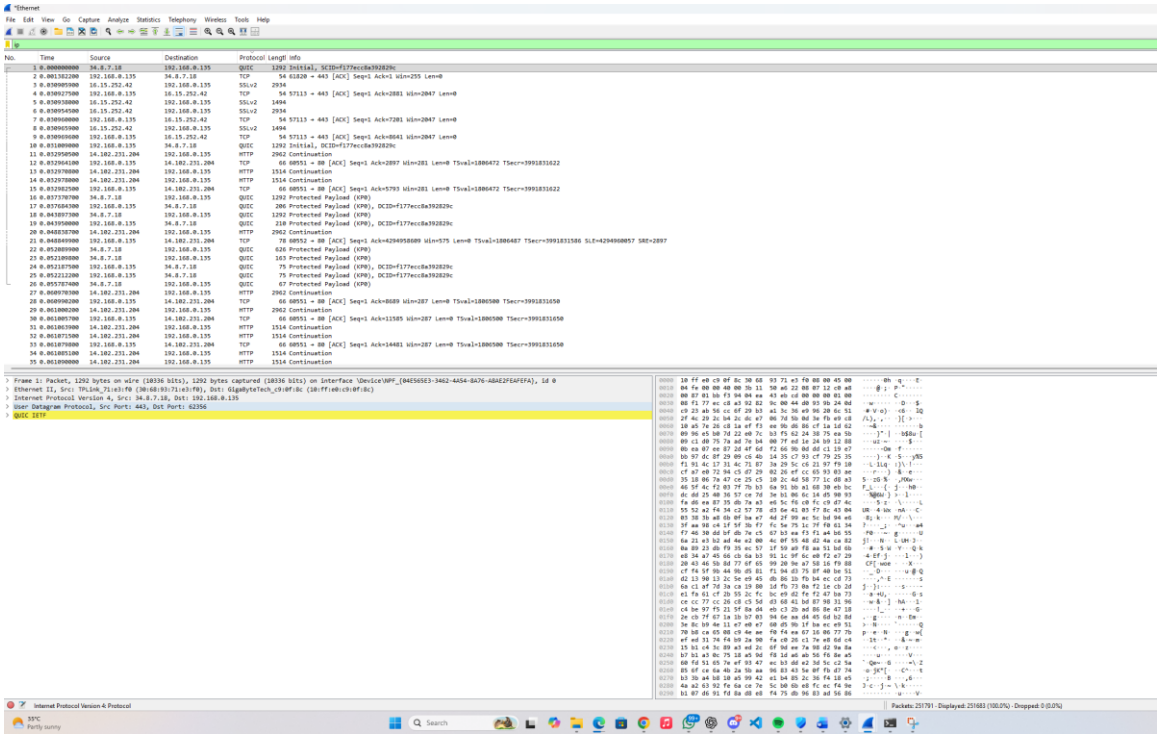
Step 4: Stop Capture

- Click the **red Stop button**
- Captured packets will remain on the screen

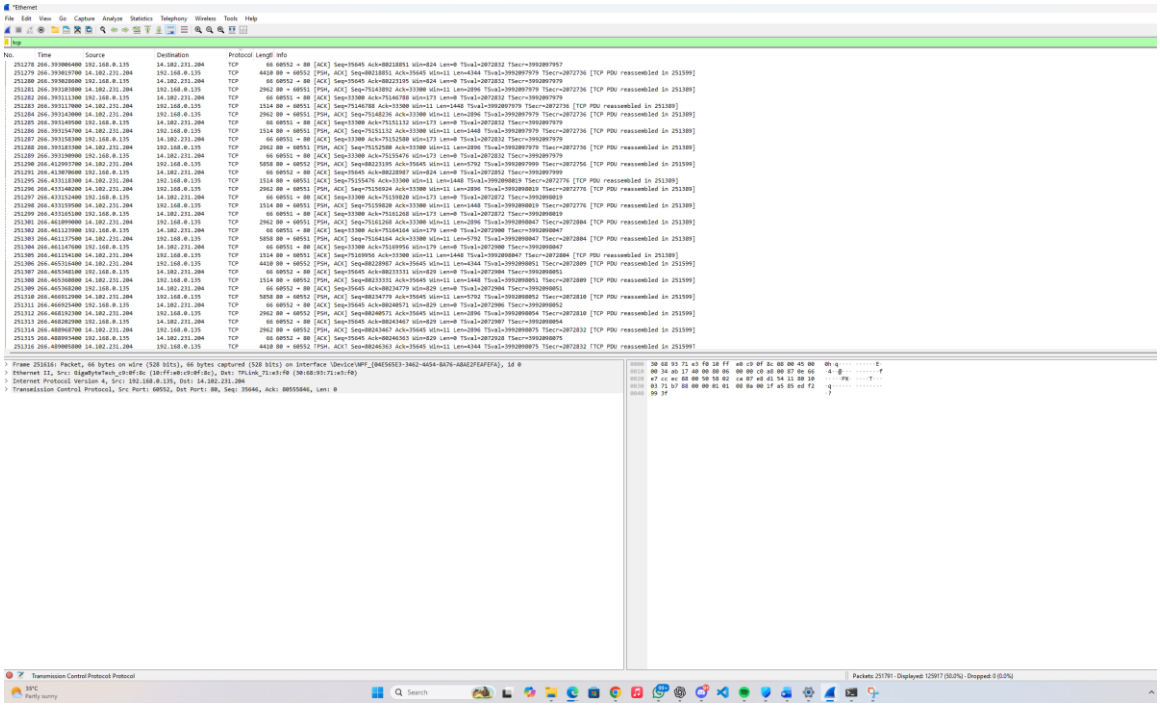
Step 5: Apply Basic Filters

Use the **Display Filter bar**:

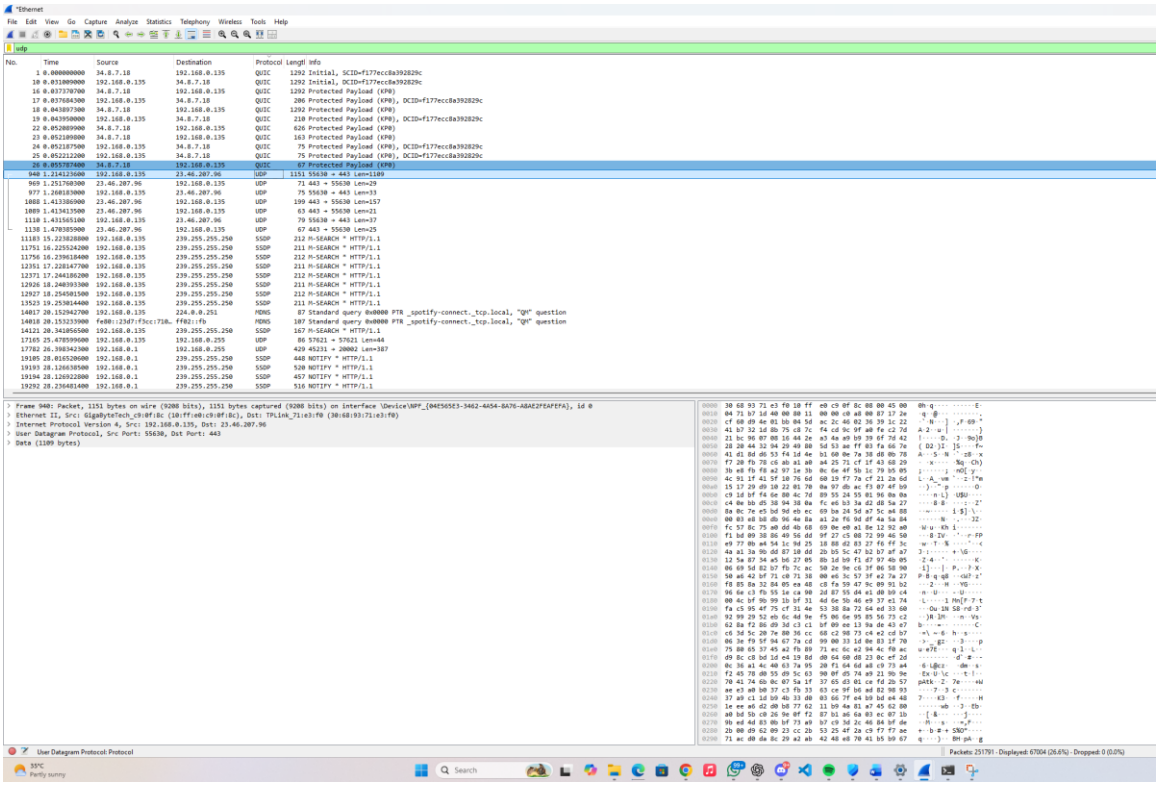
Filter	Purpose
ip	Show IP packets



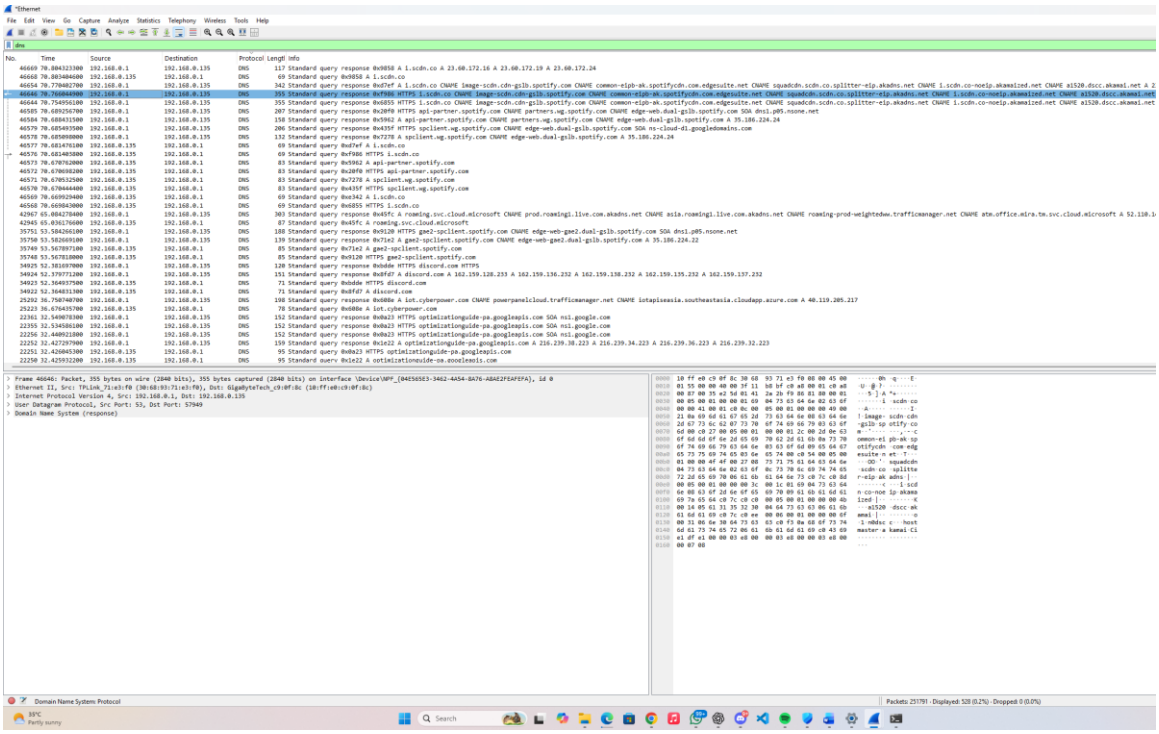
tcp Show TCP packets



udp Show UDP packets



dns Show DNS packets



icmp Show ping packets

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
173593	126.518270700	142.250.183.238	192.168.0.135	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=116 (request in 173576)
173576	126.502329300	192.168.0.135	142.250.183.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 173593)
173235	125.514017800	142.250.183.238	192.168.0.135	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=116 (request in 173234)
173234	125.496129600	192.168.0.135	142.250.183.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 173235)
165177	124.505940300	142.250.183.238	192.168.0.135	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=116 (request in 165000)
165000	124.480584300	192.168.0.135	142.250.183.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 165177)
153790	123.497078900	142.250.183.238	192.168.0.135	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=116 (request in 153595)
153595	123.478964300	192.168.0.135	142.250.183.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 153790)
35882	53.777004500	192.168.1.4	192.168.0.135	ICMP	94	Destination unreachable (Host unreachable)
30413	45.766820100	192.168.1.4	192.168.0.135	ICMP	94	Destination unreachable (Host unreachable)
28321	41.766739500	192.168.1.4	192.168.0.135	ICMP	94	Destination unreachable (Host unreachable)
26739	38.736735400	192.168.1.4	192.168.0.135	ICMP	94	Destination unreachable (Host unreachable)
26738	38.736689600	192.168.1.4	192.168.0.135	ICMP	94	Destination unreachable (Host unreachable)

```
> Frame 165177: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{04E565E3-3462-4A54-BA76-ABAE2FEAFEFA}, id 0
> Ethernet II, Src: TPLink_71:e3:f0 (30:68:93:71:e3:f0), Dst: GigaByteTech_c9:0f:8c (18:ff:e0:c9:0f:8c)
> Internet Protocol Version 4, Src: 142.250.183.238, Dst: 192.168.0.135
> Internet Control Message Protocol
```

```
0000  18 ff e0 c9 0f 8c 30 68  93 71 e3 f0 08 00 45 00   ····· 0h ·q····E·
0010  00 3c 00 00 00 00 74 01  3e a9 8e fa b7 ee c0 a8   ·<····t· >·····
0020  00 87 00 00 55 59 00 01  00 02 61 62 63 64 65 66   ····UY·· ··abcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76   ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                     wabcdefg hi
```

Internet Control Message Protocol: Protocol

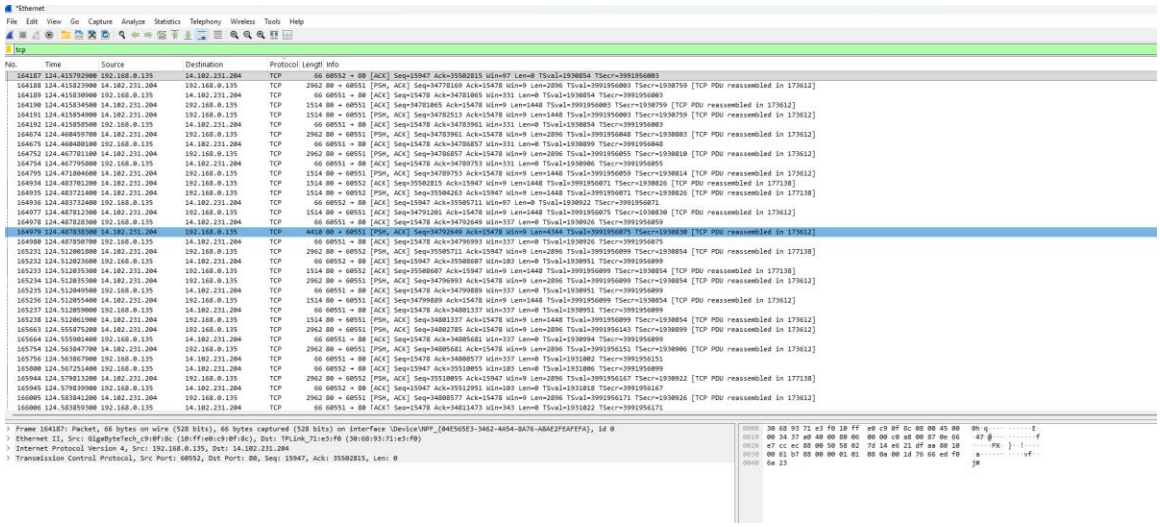
Packets: 251791 · Displayed: 13 (0.0%) · Dropped: 0 (0.0%)

35°C
Partly sunny

Search

Step 6: Analyze a Packet

- Click on any packet
- Observe:
  - **Ethernet header-**
    Source MAC Address: ec:e0:c9:0f:8c:08
    Destination MAC Address: 30:68:93:71:e3:f0
    Type: IPv4
  - **IP header-**
    Source IP Address: 192.168.0.135
    Destination IP Address: 14.102.231.204
    Protocol: TCP
  - **TCP/UDP header-**
    Source Port: 60552
    Destination Port: 80 (HTTP)
    Sequence Number: 15947
    Acknowledgment Number: 35502815
    Flags: ACK
  - **Payload data**-Length**:** 0 bytes (Len = 0 visible)



🔭 Observation

- Interface used: Ethernet (Local Area Connection)
- Protocols observed: TCP, UDP, QUIC, SSDP, MDNS, IP
- Source and destination IP addresses: 192.168.0.135, 14.102.231.204
- Packet length: **:** 66 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.